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Integrated Geological-Engineering-Economic or for Optimization Decision-Making in Unconventional Oil and Gas Horizontal Wells

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Abstract

Unconventional oil and gas resources hold immense potential and represent the primary option for alleviating the imbalance between oil and gas supply and demand. However, the high costs associated with engineering operations pose challenges for profitable development. Practical experiences have demonstrated that an integrated approach combining geological and engineering strategies is an effective means to reduce costs and enhance efficiency. Given the multitude of geological conditions and engineering parameters that influence development profit, which surpass the scope that can be comprehensively grasped by expert experience alone, the current integrated optimization decisions primarily relying on expert experience may not necessarily constitute globally optimal solutions.

In contrast, the widespread and successful applications of operations research in various sectors of the national economy, including military and engineering fields, have shown its significant potential for quantitatively solving optimal decision-making problems in complex systems. Therefore, this study leverages operations research theory to first construct production models and cost models for unconventional oil and gas resources as functions of all geological conditions and engineering construction parameters. Subsequently, a profit (= production - cost) objective function is established, thereby transforming the practical problem of optimal integrated geological-engineering decision-making into a mathematical problem of maximizing the profit objective function.

Following this, computer software is developed to determine the maximum value of the objective function and the corresponding values or matches of geological conditions and engineering parameters, enabling quantitative, scientific, and optimal decision-making in integrated geological-engineering projects. If successful, this approach will facilitate the profitable development of numerous unconventional oil and gas resources.

Introduction

Drawing on the experiences of horizontal well drilling and large-scale hydraulic fracturing technologies from the United States, after years of intensive research and practical exploration, China has developed geological and engineering technologies suited to its own geological conditions for unconventional oil and gas exploration and development. The technical feasibility of these developments has been verified. Despite the vast area and substantial total resources of unconventional oil and gas reservoirs, their resource abundance is lower than that of conventional oil and gas, making extraction challenging. Effective development necessitates the utilization of horizontal well drilling and large-scale hydraulic fracturing technologies, which contribute to high engineering operational costs. In particular, China's complex geological and surface conditions further escalate exploration and development costs, posing numerous challenges to economic feasibility.

Numerous domestic and foreign scholars have conducted extensive explorations into the integrated geological-engineering approach [1-4]. However, as will be elucidated in the subsequent review of the current research landscape in this field, the current optimization decisions for integrated geological-engineering projects predominantly rely on qualitative decision-making based on expert experience or single-factor analysis, lacking scientific, quantitative, and holistic foundations. Moreover, the prevailing optimization decisions primarily aim at maximizing production under certain investment constraints or maximizing the profitability of individual wells, rather than optimizing the overall profitability of the entire area. A holistic, quantitative solution targeting maximum profitability has yet to be formulated, which is precisely the issue this paper endeavors to explore and address.

Integrated Geological-Engineering Research

Integrated geological-engineering refers to a process that is grounded in geological studies and integrates disciplines such as reservoir engineering, drilling, and fracturing. It involves the hierarchical and dynamic optimization of drilling and fracturing engineering parameters to enhance production and efficiency [5-6]. In the context of integrated geological-engineering, "geological" encompasses the collective disciplines related to reservoir geology, including three-dimensional geological modeling and reservoir characterization, among others. "Engineering" refers to the collective disciplines related to the development process, including drilling engineering, hydraulic fracturing engineering, production engineering, and development engineering [5,7]. "Integration" denotes the achievement of interdisciplinary collaboration; whereby multiple independent and dispersed elements are combined into a workflow on a single platform.

Numerous scholars both domestically and internationally have conducted extensive explorations into integrated geological-engineering. Since 2012, the integrated geological-engineering development approach has been widely applied in unconventional oil and gas fields in North America, such as Marcellus and Eagle Ford, where activities such as scheme design and engineering parameter optimization have been carried out [8-10]. Drawing on the implementation experiences of integrated geological-engineering both domestically and internationally, Xian [11] proposed an integrated technology that combines geology, drilling engineering, fracturing engineering, production engineering, and development engineering (Fig. 1). Currently, some relatively mature comprehensive analysis platforms for integrated geological-engineering include Jewel Suite by BHGE, Petrel by Schlumberger, and Landmark by Halliburton.

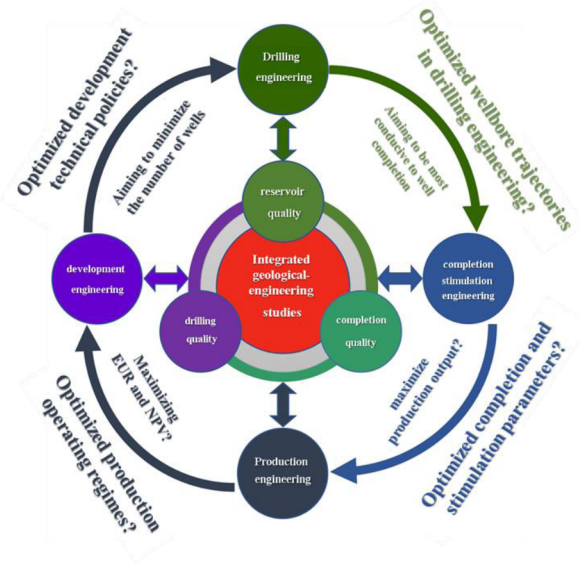


Figure 1—Geological—Engineering integrated model of unconventional oil and gas.

Integrated Geological-Engineering Research

Decision-making is a process wherein individuals, aiming to achieve specific objectives, employ scientific theories and methodologies to systematically analyze subjective and objective conditions, propose various pre-selected schemes, select the optimal one from them, and implement it. Decision-making can be categorized into qualitative and quantitative approaches. Qualitative decision-making methods include brainstorming, the Gordon method, and the Delphi method. Quantitative decision-making encompasses single-objective deterministic, single-objective risky, single-objective uncertain, multi-objective deterministic, multi-objective risky, and multi-objective uncertain types. Previous researchers have adopted a variety of decision-making methods to guide actions, mitigate risks, foster development, and enhance efficiency [12-16].

Optimization, a branch of operations research, focuses on the numerical solution of decision-making problems, possessing the potential to quantitatively address optimal decision-making in complex systems. It integrates model establishment, algorithm design, and computer technology to provide scientific and effective decision support for decision-makers. The decision-making process necessitates the support of optimization models, which, through optimization methods, solve for the optimal scheme, thereby offering scientific and effective decision support to decision-makers facing complex decision-making problems with multiple objectives and constraints.

An optimization problem is defined as the extremum problem of a function of multiple variables under certain constraints, with the mathematical model presented as follows:

$$\min f(x) \quad (1-1)$$

$$\text{s.t. } g_i(x) \leq 0, \quad i = 1, 2, \dots, m \quad (1-2)$$

$$h_j(x) = 0, \quad j = 1, 2, \dots, p \quad (1-3)$$

where $f(x)$ is the objective function; $g_i(x)$ and $h_j(x)$ are functions of the variable x , with $g_i(x) \leq 0$ representing inequality constraints and $h_j(x) = 0$ representing equality constraints; x denotes the decision variable. If for all $x \in R$, $f(x) \geq f(x^*)$, then x^* is referred to as the global optimal solution to the problem.

Optimization problems are classified into constrained and unconstrained optimization problems based on the presence or absence of constraint functions in the mathematical model. They are further categorized into linear programming, nonlinear programming, quadratic programming, integer programming, dynamic

programming, etc., based on the types of the objective function and constraint functions. Currently, algorithms for solving optimization problems are mainly divided into traditional optimization algorithms and intelligent optimization algorithms. The former includes conjugate gradient methods, Newton's methods, sequential quadratic programming, penalty function methods, etc. These algorithms typically require gradient information of the function during the solution process and are only applicable to differentiable functions, prone to getting trapped in local optima, thus exhibiting certain limitations. The latter, characterized by simple operation, high search efficiency, and strong robustness, searches from multiple points simultaneously, enhancing the search efficiency of the algorithm. They do not require the calculation of the function's gradient information and are suitable for solving complex optimization problems. Commonly used optimization algorithms include the Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Differential Evolution (DE) algorithm, among others.

Methods

In order to enhance the overall exploration and development efficiency of unconventional oil and gas resources, and to provide holistic and quantitative solutions for horizontal well patterns, well spacings, horizontal section lengths, fracturing stage counts/scales, and proppant volumes, this paper conducts research on various aspects including production models, cost models, benefit objective functions, optimization decision-making algorithms, decision software systems, and their applications. These research components are intricately interconnected.

Numerical Simulation Model for Multi-stage Fractured Single Wells

Taking the P1 shale gas block as an example, based on the geological characteristics of the P1 block, well test interpretation data, and the historical matching results of numerical simulation, this paper utilizes CMG software to establish a mathematical model for the flow of shale gas in horizontal wells. The model accounts for gas-water two-phase flow, a multiple-porosity medium model, non-Darcy flow, horizontal well friction, and adsorption-desorption effects. The model dimensions are 6000 m in length and 800 m in width, with a grid system of $300 \times 40 \times 5$, totaling five sub-layers. The established reservoir model is a rectangular grid system centered around the horizontal well. In the reservoir model, the horizontal well is positioned in sub-layer ①, with a production cycle set at 15 years. Hydraulic fractures propagate along the direction of the maximum principal stress. The numerical simulation employs a local grid refinement method to accurately simulate fluid flow within the hydraulic fractures. Input parameters for the conceptual model are determined based on core data, historical matching results, field data, and PVT data.

The relative permeability curves required for the simulation are provided by the results after historical matching. Fig. 2(a) displays a top view of a multi-stage fractured horizontal well. The pressure distribution after 15 years of production indicates that the pressure drawdown gradually propagates from the wellbore into the distant formation (Fig. 2(b)). Additionally, the pressure drawdown predominantly occurs near the fractures, confirming that hydraulic fracturing technology can enhance shale gas production.

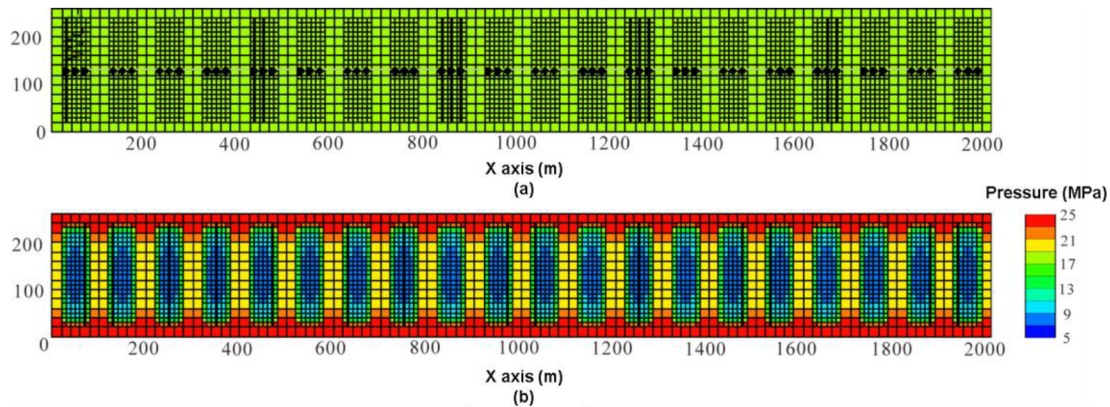


Figure 2—Establish a numerical model for multi-stage fractured horizontal wells in unconventional oil and gas reservoirs. (a) Numerical simulation model of multistage fracturing horizontal well; (b) Pressure distribution of numerical simulation model after 15 years of production

Establish the relationship between production and engineering costs

Since the variables in the production model differ from those in the fracturing cost model—the former being fracture half-length and conductivity, while the latter being fracturing fluid volume and proppant mass—it is necessary to establish a quantitative relationship between fracturing parameters and fracture parameters, thereby bridging the variables in production and fracturing cost. Commonly used methods for obtaining this quantitative relationship include micro seismic methods, empirical formula methods, and fracturing simulation techniques. Given that micro seismic technology can only roughly monitor the fracture propagation direction and is unable to accurately evaluate fracture geometry, propped fracture height, and length, the quantitative relationship between fracturing and fracture parameters established through micro seismic monitoring cannot be utilized in the optimization process. The empirical formula method determines the functional relationship between fracturing fluid cost and fracture parameters based on the actual fracturing cost and fracture parameter fitting relationship in the study area. However, empirical formulas require a substantial amount of cost data and actual fracture parameters, which are often difficult to accurately calibrate. Fracturing simulation software considers the inter-coupling and penetration relationships among multiple disciplines, including fluid mechanics, solid mechanics, fracture mechanics, and thermodynamics. It can simulate the complex fracture networks formed during the volume fracturing process of unconventional oil and gas reservoirs under given fracturing fluid and proppant conditions [17], thereby obtaining the relationship between fracturing and fracture parameters. Currently, reservoir fracturing simulation and design software have formed a relatively complete system, encompassing fracturing simulation, fracturing optimization design, as well as real-time construction monitoring and analysis. However, the computational cycle of fracturing simulation software is lengthy, and hundreds of runs may be required during the optimization process, making it difficult to achieve rapid decision-making. Therefore, the consideration of establishing a fracturing surrogate model using machine learning is warranted.

Gaussian Process Regression (GPR) is a non-parametric supervised learning model based on Bayesian theory, with a rigorous statistical theoretical foundation. It can address complex regression problems and is widely applied in the industrial sector [18–21]. The Gaussian model is determined by the mean and covariance function of the input data. Assuming that the relationship f between the input data and output values follows a Gaussian process [19]:

$$f \sim GP(m, k) \quad (2-1)$$

For a training set, the regression model is expressed as:

$$y = f(x) + \varepsilon \quad \varepsilon \sim N(0, \sigma^2) \quad (2-2)$$

where x is the input vector, f is the predictive function, y is the output target value, and ε represents noise.

For an input vector $x = (x_1, x_2, \dots, x_n)$, assuming $f(x_i)$ follows a Gaussian distribution, then $f(x)$ follows a multivariate Gaussian process:

$$f(x) \sim GP(m(x), k(x, x')) \quad (2-3)$$

$$m(x) = E[f(x)] \quad (2-4)$$

$$k(x, x') = E[(f(x) - m(x))(f(x') - m(x')))] \quad (2-5)$$

where $m(x)$ and $k(x, x')$ are the mean and covariance of $f(x)$, respectively.

By processing the data such that $m(x) = 0$, the prior distribution of the observed values y is:

$$y \sim N[0, K(x, x) + \delta_n^2 I] \quad (2-6)$$

The training set y and the predictive set f_* follow a Gaussian distribution, with the joint prior distribution given by:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim N\left(0, \begin{bmatrix} K(x, x) + \delta_n^2 I_n & K(x, x_*) \\ K(x_*, x) & k(x_*, x_*) \end{bmatrix}\right) \quad (2-7)$$

where $K(x, x) = K_n = (k_{ij})$ is an $n \times n$ symmetric positive definite covariance matrix; $K(x, x_*) = K(x_*, x)^T$ is the covariance matrix between the training data and test data; $K(x_*, x_*)$ is the covariance of the test data x_* ; and I_n is the n -dimensional identity matrix.

The posterior distribution of f_* is:

$$f_* | X, y, x_* \sim N(\bar{f}_*, \text{cov}(f_*)) \quad (2-8)$$

$$\bar{f}_* = K(x_*, x) [K(x, x) + \sigma_n^2 I_n]^{-1} y \quad (2-9)$$

$$\text{cov}(f_*) = k(x_*, x_*) - K(x_*, x) \times [K(x, x) + \sigma_n^2 I_n]^{-1} K(x, x_*) \quad (2-10)$$

where $\bar{f}_*$ and $\text{cov}(f_*)$ are the mean and covariance of f_* , respectively.

The kernel function (also known as the covariance function) influences the accuracy of the GPR model. Common kernel functions include squared exponential, exponential, Matern 3/2, Matern 5/2, rational quadratic, and ARD squared exponential. For instance, the formula for the squared exponential kernel is:

$$k(x_i, x_j | \theta) = \sigma_f^2 \exp\left[-\frac{1}{2} \frac{(x_i - x_j)^T (x_i - x_j)}{\sigma_l^2}\right] \quad (2-11)$$

where x_i and x_j are the input data for training, and σ_f and σ_l are the hyperparameters of the kernel function.

Through Gaussian Process Regression (GPR) analysis, a quantitative relationship is established between the fracturing fluid volume, proppant mass, fracture half-length, and fracture conductivity. The range of each designed variable is set as wide as possible to cover the spectrum of fracture half-lengths and conductivities within the production surrogate model. By running the established fracturing simulation model, corresponding fracture half-lengths and conductivities can be obtained. During the calculation process, the values of non-variable parameters are consistent with those in the conceptual model of the fracturing simulation. Fig. 3 displays a relationship diagram obtained from the experimental design, illustrating the correlations among fracture half-length, conductivity, fracturing fluid volume, and proppant concentration.

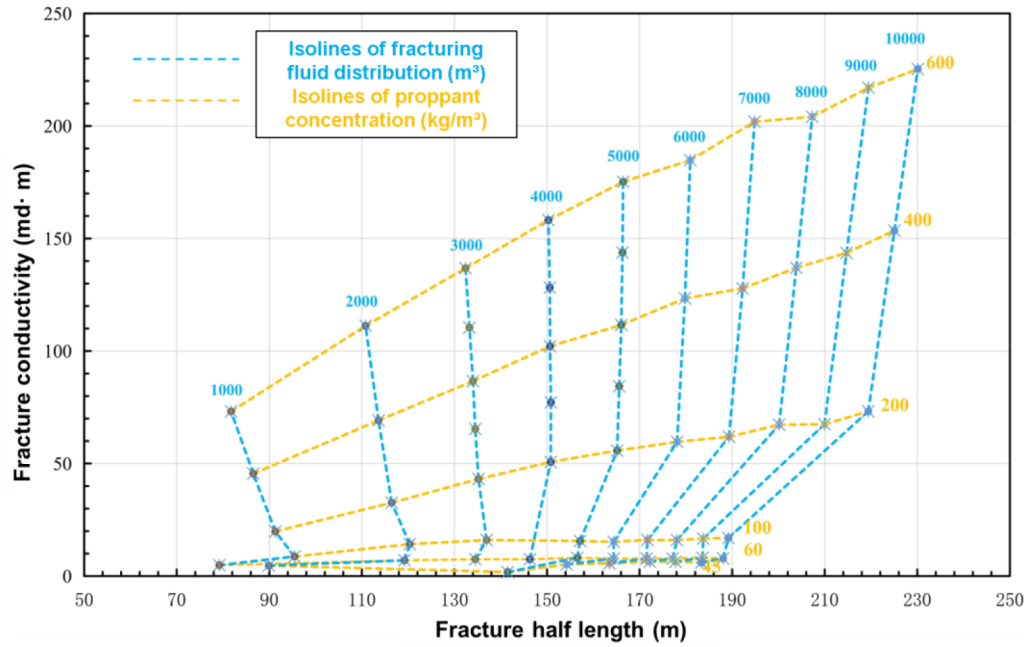


Figure 3—Relationship diagram of fracture half-length, fracture conductivity, fracturing fluid volume and proppant concentration.

Optimization of the Objective Function

To enhance the profitability of individual wells and the entire block, the economic Net Present Value (NPV) of the study area is set as the objective function to optimize the engineering parameters of shale gas horizontal wells. This is divided into a single-well NPV function and a block NPV function.

Single-Well NPV Model. The NPV consists of two parts: cash inflow and cash outflow. The single-well NPV is derived by subtracting the operating costs, taxes, drilling costs, and fracturing costs from the single-well revenue. The NPV model comprehensively considers these cost factors to provide a holistic assessment of the profitability of a single well. The formula is as follows:

$$NPV_{single} = \sum_{t=1}^T \left[\frac{1}{(1+b)^t} (P_0 Q_{gas} - C_{operating} - C_{tax}) \right] - f_{drill} - f_{frac} \quad (2-12)$$

where: NPV_{single} is the NPV of a single well, in yuan; $t = 1, 2, 3, \dots, T$ represents time in years; b is the discount rate, %; P_0 is the natural gas price, in yuan/m³; Q_{gas} is the annual gas production, in m³/year; f_{drill} is the drilling cost, in yuan; f_{frac} is the fracturing cost, in yuan; $C_{operating}$ is the annual operating cost, in yuan/year; C_{tax} is the sales tax and resource tax, in yuan/m³.

By analyzing the composition of operating costs and taxes in the study area, formulas for operating costs, sales tax, and resource tax are established as follows:

$$C_{operating} = 3 \times 10^6 + 0.3 Q_{gas} \quad (2-13)$$

$$C_{tax} = 0.006 Q_{gas} \quad (2-14)$$

Block NPV Model. The block NPV is derived by subtracting the total operating costs, taxes, drilling costs, fracturing costs, and surface costs from the revenue of all gas wells. The NPV model considers all revenues and investments to provide the most accurate NPV calculation. The formula is as follows:

$$NPV_w = \sum_{j=1}^N \sum_{t=1}^T \left[\frac{1}{(1+b)^t} (P_0 Q_{gas} - C_{operating} - C_{tax}) \right] - \sum_{j=1}^N (f_{drill} + f_{frac} + C_{ground}) - \sum_{i=1}^m C_{platform} \quad (2-15)$$

where: NPV_w is the total NPV of the entire block, in yuan; $j = 1, 2, 3, \dots, N$ represents the number of wells; $i = 1, 2, 3, \dots, m$ represents the number of platforms; $C_{platform}$ is the cost of a single drilling platform, in yuan; C_{ground} is the surface cost, in yuan.

GPR-BPGA-CCIMS

The Gaussian Process Regression - Bilevel Programming Genetic Algorithm - Critical Circle Intersection Momentum Shift (GPR-BPGA-CCIMS) method decomposes the optimization problem into hierarchical levels. However, the key difference lies in that the GPR-BPGA-CCIMS method optimizes both the location and number of drilling platforms simultaneously. The optimization steps are as follows:

1. Establishing a Geological Model: Based on core, logging, and seismic data from the study area, porosity models, gas saturation models, formation pressure models, and structural models are constructed.
2. Establishing Production, Cost, and Profitability Models: Machine learning and numerical simulation are employed to establish production surrogate models. By analyzing the relationship between actual costs and engineering parameters in the study area, functions linking costs to engineering parameters are derived. Ultimately, the practical problem of integrated geological-engineering optimal decision-making is transformed into a mathematical problem of maximizing a profitability objective function, based on the geological, production, and cost models.
3. Proposing the GPR-BPGA-CCIMS Algorithm to Optimize the Block's NPV Function: The results of the inner-level optimization function (the maximum NPV for each column) serve as constraints for the outer-level optimization function. The BPGA algorithm, which combines Bilevel Programming (BP) and Genetic Algorithm (GA), is referred to as a bilevel genetic algorithm. The outer-level optimization variables are x_f , F_c and m_f , while the inner-level optimization variables are X_A , X_B , and L . The outer-level optimization algorithm first generates initial values for x_f , F_c and m_f . Based on the given outer-level x_f it calculates the abscissas and burial depths of each row of wells from the bottom-left to the top-right corner, and determines the optimal range for L based on the drilling rig's ultimate bearing capacity and burial depth. The inner-level optimization algorithm generates initial values for L and N for each column of wells and calculates the NPV for each column. The inner-level GA iterates multiple times to generate the optimal L and N for each column. Subsequently, the Critical Circle Intersection (CCI) method and the Momentum Shift (MS) method are employed to merge and shift the drilling platform locations, thereby obtaining the optimal drilling platform locations and number for that iteration. Finally, the NPV for the entire area is calculated. The outer-level optimization algorithm iterates multiple times. When the maximum number of iterations is reached, the optimal values for x_f , F_c , m_f , X_A , L , D_A , X_B , N , the number of drilling platforms, and their locations for the entire area are output.

Bilevel Programming. Bilevel Programming (BP) is an important application problem within mathematical programming [22]. If the result of a nonlinear programming problem is utilized as a constraint condition for another optimization problem, then this optimization problem is referred to as bilevel programming [23-24]. The model for bilevel programming is as follows:

$$\min F(x, y) \quad (2-16)$$

$$x, y \text{ s.t. } \begin{cases} G(x, y) \leq 0 \\ H(x, y) = 0 \\ \min f(x, y) \end{cases} \quad (2-17)$$

where: $F(x, y)$ is the objective function of the outer level; x, y are the decision variables; $G(x, y) \leq 0$ represents the inequality constraints; $H(x, y) = 0$ represents the equality constraints; $\min f(x, y)$ is the objective function of the inner level, which serves as a constraint condition for the outer-level objective function.

(2) Critical Circle Intersection Method - Momentum Shift Method

The objective of the Critical Circle Intersection (CCI) method is to merge drilling platforms, while the Momentum Shift (MS) method aims to obtain the optimal drilling locations and number by shifting the positions of the merged drilling platforms.

Fig.4 illustrates the principle of the CCI method. Firstly, an initial drilling platform is assigned to each well, followed by the merging of drilling platforms. Since the planar distance between the drilling platform and the horizontal well must be greater than 343 m, the drilling platform can only be located on or outside the critical circle. The critical circle is defined as a circle with a radius of 343 m centered at the target point A (A1 and A2 in Fig.4). When the drilling platform is located outside the critical circle (P1 in Fig.4), there exists at least one point (P2) on the critical circle such that $(P1A1 + P1A2) > (P2A1 + P2A2)$. Therefore, points outside the critical circle are not optimal. When the drilling platform is located on the critical circle, the intersection point (C4) of the two critical circles achieves the shortest distance between the platform and the target points. When there is no intersection between the two critical circles, the intersection point (C2 or C3) of the lower critical circle with the horizontal line or the lower intersection point (C1) of the middle vertical line with the adjacent critical circle is the desired point. The costs for each platform and corresponding wells are calculated one by one, and the point with the lowest drilling cost is selected as the initial platform location through comparison.

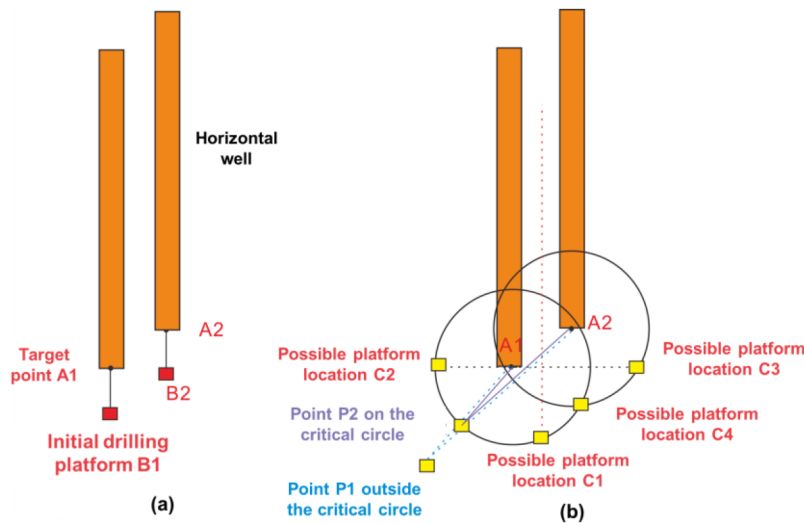


Figure 4—Critical Circle Intersection Method - Momentum Shift Method (a) Initial drilling platform location; (b) Consolidation of drilling platform using the Critical Circle Intersection (CCI) method.

After the platforms are merged, it is necessary to relocate the platforms to their optimal positions to further reduce drilling costs. The cost of a drilling platform is related to the distance between the platform and the wells, and the cost increases with the distance. Therefore, the objective of the Momentum Shift (MS) method is to position the drilling platforms as close as possible to the horizontal wells to reduce drilling costs. The direction of platform movement is crucial. In this paper, a simple yet effective method is adopted, which involves calculating the resultant vector of the platform and the connected wells, and then moving the drilling platform along this resultant vector to simultaneously consider the influence of all wells connected to the platform. To ensure that the cost is reduced after moving the drilling platform, the costs before and after the movement are calculated separately. The platform is only moved if the cost decreases.

The formula for the MS method is as follows:

$$(x_2, y_2) = (x_1, y_1) + 0.1 \times (x_s, y_s) \quad (2-18)$$

where: (x_1, y_1) and (x_2, y_2) represent the positions of the platform before and after the drift, respectively; 0.1 is the movement step size; (x_s, y_s) is the resultant vector determined by the positions of the horizontal well and the connected drilling platforms.

After the platform positions are arranged, the NPV for each column is calculated by invoking geological parameters, and the NPV value for the entire block corresponding to this iteration is obtained. Subsequently, the outer-level optimization algorithm iterates to generate new fracture half-lengths, and the aforementioned iterative process is repeated. Finally, the maximum NPV for the entire block, along with the optimal drilling layout and fracturing design scheme, is obtained.

Application and Results

The proposed optimization algorithms were applied to Block P1. Currently, the actual average horizontal section length, well count, well spacing, cluster spacing, and conductivity range in Block P1 are 1516 m, 29 wells, 500 m, 10-30 m, and 10 md·m, respectively. The ranges for fracturing fluid volume per meter and proppant volume per meter are 20-35 m³ and 0.7-1.1 m³, respectively. The NPV and recovery factor are 7.4×10^8 yuan and 11%. Using the GPR-MLCGA-ES optimization algorithm, under the condition of fixed drilling platforms, the optimal NPV values obtained for the entire block after well placement using a 70 rig and a combined 50-70 rig are 10.5×10^8 yuan and 10.6×10^8 yuan, respectively. Using the GPR-BPGA-CCIMS optimization algorithm, under the condition of non-fixed drilling platforms, the optimal NPV values obtained for the entire block after well placement using a 70 rig and a combined 50-70 rig are 11.6×10^8 yuan and 12.4×10^8 yuan, respectively.

The optimal ranges for well count, well spacing, cluster spacing, and conductivity across the entire block are 30-34 wells, 260-324 m, 20-22 m, and 13-15 md·m, respectively. The optimal ranges for fracturing fluid volume per meter and proppant volume per meter across the entire block are 28-42 m³ and 1.5-2.6 m³, respectively. The optimal range for the average horizontal section length across the entire block is 2653-2737 m. Based on the current geological conditions, drilling technology, fracturing technology, and cost conditions, the optimal well spacing for the horizontal well pattern in Block P1 is 35%-48% shorter than the deployed well spacing, the optimal horizontal section length is increased by 52%-149%, the optimal cluster spacing remains unchanged, the optimal fracture conductivity is improved by 30%-50%, the optimal sand intensity is increased by 66%-188%, and the optimal fracturing fluid volume per meter is increased by 20%-40%. The cumulative production calculated by different optimization schemes ranges from 51×10^8 to 53×10^8 m³, with a recovery factor of 19%-20%. The optimized cumulative gas production is 64%-70% higher than that of the actual scheme, the recovery factor is increased by 8%, the benefits are improved by 35%-62%, and the NPV optimized under the condition of non-fixed drilling platforms is higher than that optimized under the condition of fixed drilling platforms.

The NPV optimized using a combined rig is higher than that optimized using a single rig. The optimization schemes significantly improve economic benefits, recovery factors, and cumulative gas production, demonstrating the immense potential of "cost reduction and efficiency enhancement" through comprehensive, quantitative geological-engineering integration, which warrants intensive research and widespread application. The trend in development schemes is to increase the proppant volume, reduce well spacing, and increase horizontal section length, which is consistent with previous optimization trends [25].

The variation of NPV with the number of iterations was plotted to analyze the convergence and optimization effects of different algorithms (Fig. 5). The GPR-MLCGA-MS algorithm outperforms the GA algorithm within a limited number of iterations. The GA algorithm initially converges to a well placement scheme and ignores other schemes during rapid convergence, thus easily falling into local optima. The GPR-MLCGA-MS algorithm exhibits better global search capabilities and achieves higher NPV values.

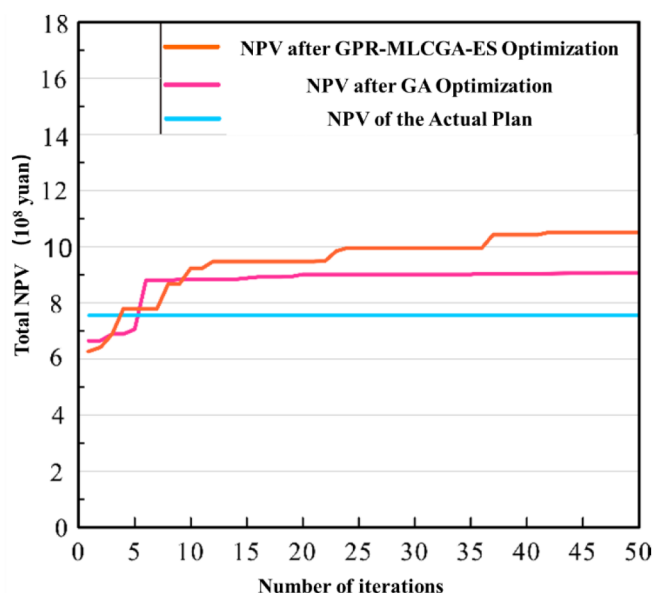


Figure 5—Field NPV iteration diagram of different methods.

Fig. 6 displays the pressure field maps after 15 years of production for both the original well placement scheme and the optimized well placement scheme. The pressure field maps before and after optimization also indicate that, compared to the pre-optimization scenario, the formation pressure across the entire block is generally reduced after optimization, more reserves are produced, and production is increased. After optimization, the well spacing is decreased, the horizontal section length is increased, and the well count is increased.

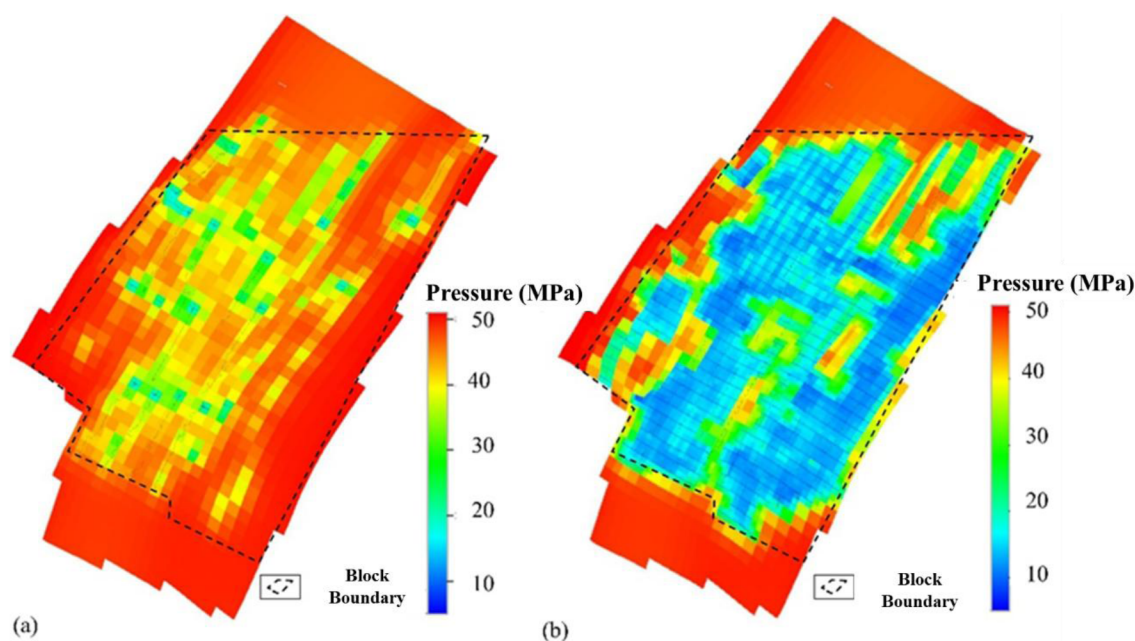


Figure 6—Pressure field maps of P1 block (a) Pressure field map of P1 block after 15 years of production of original well placement scheme; (b) Pressure field map of P1 block after 15 years of production of optimized well placement scheme (constraint conditions are fixed drilling platform and 70 rig).

In this paper, an Intel Xeon Gold 6240 processor was used to run the optimization algorithm for optimizing the engineering parameters across the entire block. Under 32-core computation, the optimization takes 2 days to complete. If traditional numerical simulation coupled with optimization algorithms were

used for engineering parameter optimization, it would take at least two months to run, even with an increase in the number of cores of the reservoir simulator. The solution time would still constrain the application of traditional optimization in large-scale practical reservoirs. Using the optimization algorithm proposed in this paper, it takes 8 hours to complete one optimization for the entire block, significantly improving the solution efficiency.

From the perspective of optimization effects, the optimal benefits and production obtained by the algorithm proposed in this paper are higher than those of the actual development scheme. In terms of optimization efficiency, the computational efficiency of the algorithm proposed in this paper is far higher than that of traditional methods. The proposed geological-engineering-economic integrated optimization algorithm achieves rapid optimization of engineering parameters across the entire block, solves large-scale, multi-variable, and multi-constraint optimization problems, provides optimal fracturing design, horizontal well, and drilling platform placement schemes for the entire block, and expands the optimization results of previous studies.

It is worth noting that, in practice, due to the uncertainties involved in various conditions and parameters, quantitative, global optimization may need to be conducted iteratively and dynamically adjusted according to actual situations. For example, during the production of unconventional oil and gas, factors such as geological conditions, engineering parameters, policies, and market conditions may change, thereby affecting the implementation effects of the optimization scheme. Therefore, dynamic adjustments need to be made according to actual situations to achieve better economic benefits.

Conclusion

This paper, based on optimization theory, constructs an unconventional oil and gas production model and a cost model that take into account both geological conditions and engineering construction parameters. Furthermore, it establishes a benefit objective function (defined as production multiplied by unit price minus cost). By doing so, the practical problem of making optimal decisions through the integration of geology and engineering is transformed into a mathematical problem of seeking the maximum value of the benefit objective function. An optimization algorithm is proposed, and computer software is developed to solve for the maximum value of the objective function as well as the corresponding values or optimal matching of geological conditions and engineering parameters. This approach enables the realization of a holistic, quantitative, scientific, and optimal decision-making process that integrates geology, engineering, and economics. The main achievements obtained are as follows:

1. Numerical simulation software was employed to develop a horizontal well fracturing model for Block P1. A Gaussian Process Regression (GPR) model was utilized to establish a quantitative relationship between production and geological conditions as well as engineering parameters. The results indicated that the GPR model exhibited high prediction accuracy, enabling the rapid establishment of single-well and block production models that closely approximate the accuracy of numerical simulation models, thereby effectively enhancing the optimization speed. Additionally, a quantitative relationship between unconventional oil and gas production and costs was established.
2. Based on the constructed geological, production, and cost models, a block benefit objective function was formulated, transforming the practical problem of optimal decision-making through the integration of geology and engineering into a mathematical problem of seeking the maximum value of the overall benefit objective function. A GPR-BPGA-CCIMS algorithm was proposed to solve the benefit model under various constraints. The proposed integrated optimization algorithm for geology, engineering, and economics addressed the challenges of multiple decision variables, numerous constraints, large computational volumes, and lengthy processing times, enabling rapid optimization to obtain the maximum benefit and corresponding matching schemes for geological, drilling, and fracturing engineering parameters. This algorithm effectively overcame the limitations

of intelligent optimization algorithms, improving optimization effectiveness and demonstrating its validity and stability in solving complex large-scale optimization problems.

3. The optimization results revealed that, under the currently recognized geological conditions (models), drilling technologies, fracturing technologies, and cost constraints, the optimal well spacing for horizontal well patterns in Block P1 ranges from 260 to 324 m, the optimal fracture half-length from 80 to 120 m, the optimal average horizontal section length approximately from 2653 to 2737 m, the optimal cluster spacing around 20 m, the optimal fracture conductivity from 13 to 15 md·m, the sand addition intensity up to 1.5 to 2.6 m³/m, and the fracturing fluid volume per meter from 28 to 42 m³/m. Compared with the actual well deployment plan, the optimized horizontal section length increased, along with the fracturing fluid volume and sand addition volume, providing theoretical support for the recent engineering practice trend of increasing horizontal section lengths and fracturing intensities.
4. During the development of unconventional oil and gas, the integrated approach of geology, engineering, and economics indeed holds significant potential for "cost reduction and efficiency enhancement." However, fully realizing this potential relies on the integration of multidisciplinary knowledge, including geology, drilling, fracturing, gas production, and economic evaluation. Specifically, the complex and vast engineering problem of optimizing the combination of geological conditions and construction parameters affecting the development efficiency of unconventional oil and gas, which exceeds the qualitative decision-making capabilities of experts relying solely on experience, needs to be transformed into a mathematical problem of seeking the maximum value of the benefit objective function using optimization theory. This approach provides a holistic, quantitative, and scientific decision-making scheme for achieving geology-engineering integration. Compared with the original well deployment and construction plans, the optimized well deployment and construction plans obtained in this study demonstrated substantial efficiency improvements. For instance, the optimized cumulative gas production in the study target area (Block P1) increased by 64% to 70% compared with the actual plan, the recovery factor increased by 8%, and the benefit increased by 35% to 62%. This technology also holds potential for further optimizing the target azimuth and in-situ stress angle of horizontal wells, the horizontal well traversal horizons, and the configuration of surface supporting engineering facilities.
5. The optimal horizontal well pattern (well spacing, horizontal section length) is not only related to geological conditions (porosity, gas saturation, formation pressure, etc.) but also closely tied to advancements in drilling and fracturing technologies (progress in drilling technologies will reduce well spacing, while progress in fracturing technologies will increase well spacing). Geology-engineering integrated optimal decision-making requires the quantitative optimization of well pattern deployment schemes based on the construction of the benefit objective function for the entire area and consideration of various constraints (such as fault development distribution, topography, water source conditions, ecological zones, and surface mineral rights areas). This should be the development direction for quantitative optimal decision-making in the integration of geology, engineering, and economics.

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